

Aditya Aryan

Bachelor of Statistics student

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📍 Kolkata, West Bengal, India

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Profile

- Bachelor of Statistics student at the Indian Statistical Institute, Kolkata, interested in theoretical and applied statistics and machine learning. Experience includes EM convergence research, applied statistical modelling, and medical vision-language learning.

Education

Bachelor of Statistics (B.Stat), Indian Statistical Institute, Statistics
• Expected graduation: 2027.

Kolkata, West Bengal
July 2023 – 2027

Secondary and Secondary Education, Kendriya Vidyalaya No. 2, Jalahalli East, Physics, Chemistry, and Mathematics
• Class XII: 95%.
• Class X: 98.8%.

Bengaluru, Karnataka
Apr 2014 – Mar 2023

Experience

University of Toronto, Student Researcher
Research with Prof. Xin Bing on expectation-maximisation for Gaussian mixture-type models.
• Study population and sample EM for Gaussian mixture-type models with Prof. Xin Bing.
• Analyse update maps, fixed points, and local contraction, including the effects of component imbalance and sampling error.

Remote
Apr 2026 – present
6 months

Ranial Systems, Data Science Intern
Statistical and machine-learning methods for energy systems.
• Develop statistical and machine-learning workflows for battery-health modelling and probabilistic forecasting in energy systems.
• Build evaluation pipelines that distinguish prediction, risk ranking, and decision performance, with attention to time and group structure.
• Investigate forecasting and battery-constrained decision methods for electricity-market applications.

Remote
May 2026 – present
5 months

Mercity AI, Machine Learning Research Intern
Compact multimodal modelling for medical image caption generation.
• Worked on a Q-Former interface between medSigLIP and frozen Qwen-2.5-7B for medical caption generation, compressing 256 visual tokens into 32 queries.
• Explored supervised and reward-based training for multimodal models.

Bengaluru
Apr 2025 – Aug 2025
5 months

Projects

James–Stein Shrinkage and Empirical Bayes Risk

Supervised by Dr. Ayanendranath Basu, Indian Statistical Institute.

- Studied classical shrinkage risk and empirical Bayes extensions; compared Monte Carlo risk with SURE under specified normal-means models.

Optimal Stopping in a Finite Reward-Sampling Game

Supervised by Dr. Arnab Chakraborty, Indian Statistical Institute.

- Formulated a Bellman recursion and implemented the optimal policy in C++; checked the reported instance by exhaustive enumeration.
- For $N = 20$, $C = 5$, reward value $V = 5$, and unit opening cost, obtained expected gain 7.6876.

Copula Modelling of Air-Pollution Episodes

Supervised by Dr. Shyamal Krishna De, Indian Statistical Institute.

- Compared six copula families for pollution-episode duration and cumulative severity, interpreting tail dependence alongside the variables' structural relationship.

Nonlinear-MLP: Controlled Experiments on Neural-Network Nonlinearity

Independent research software in development.

- Built an experimental framework for activation replacement, learned gates, and control comparisons; comparative evaluation ongoing.

BioStat-PO: Selecting Analysis Pipelines for Causal Survival Inference

Independent research software in development.

- Developed a Python–R framework for analysis-pipeline selection under missingness and censoring; synthetic-data development pilots available.

Skills

Statistical methods: Estimation and decision theory, Probabilistic modelling, Dependence modelling, Simulation-based evaluation

Machine learning: PyTorch experiments, Neural networks, Vision-language modelling, Supervised and reward-based training workflows

Programming: Python, R, C++ for dynamic programming

Libraries: PyTorch, NumPy, SciPy, pandas, scikit-learn, statsmodels, Hugging Face tools

Coursework

- Probability Theory; Real Analysis; Linear Algebra; Stochastic Processes; Statistical Methods
- Parametric Inference; Sample Surveys; Design and Analysis of Algorithms (in progress, Fall 2026)

Honors and Awards

Indian Statistical Institute Entrance Examination

All India Rank 90

2023

Indian National Mathematical Olympiad

Merit list

2022

Indian Olympiad Qualifier in Mathematics

Karnataka top 20

2022

National Talent Search Examination Scholarship

National scholarship recipient; Stage I Rank 72

2021